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Chapter 1

INTRODUCTION

Snatch theft remains one of the most pervasive street crimes in developing nations, particularly in urban India. It occurs in seconds, leaving victims with no time to react, seek help, or alert emergency contacts. As cities grow and streets become busier, the risk to students, working professionals, and daily commuters — especially women — continues to rise. Despite increased surveillance infrastructure such as CCTV systems and police patrolling, these measures remain reactive rather than proactive: they record incidents but do not prevent them or alert victims in real time.

The rapid proliferation of IoT devices and low-cost microcontroller platforms such as Arduino has opened new avenues for building affordable, deployable personal safety systems. Accelerometers, GSM modules, GPS receivers, and Bluetooth communication chips, once exclusive to high-end industrial applications, are now available for under Rs. 500 each. This democratization of hardware enables engineering students and researchers to build working prototypes addressing real social problems.

SmartGuard leverages these advances to create a wearable, autonomous theft detection and alert system. By combining G-force threshold detection via the MPU6050 accelerometer with Bluetooth-based proximity monitoring and real-time WhatsApp alert delivery, SmartGuard provides a two-layer protection mechanism: it detects violent snatches instantly and also monitors if a bag is inadvertently left behind.

1.1 Brief history of Theft Detection Systems

Early theft detection systems relied entirely on passive surveillance. CCTV cameras were introduced in public spaces in the 1970s and have since become ubiquitous. While effective at recording events for post-incident investigation, they offer no real-time alert capability to victims. The introduction of GPS vehicle tracking in the 1990s marked the first application of real-time location monitoring for stolen property, but these systems were designed for vehicles and required expensive infrastructure.

The advent of smartphones introduced personal tracking applications such as Apple Find My and Google's Find My Device. These platforms use crowdsourced networks of devices to locate lost items. Apple's AirTag, launched in 2021, brought this concept to a consumer-grade personal item tracker. However, AirTag is entirely passive — it has no alarm, no snatch detection capability, and no automatic emergency alert mechanism.

Research into accelerometer-based detection grew alongside the fitness wearable industry the 2010s. Devices like Fitbit and Apple Watch demonstrated that consumer-grade accelerometers could reliably detect human movement, falls, and physical activities. Applying this technology to theft detection — specifically detecting the distinctive G-force signature of a violent bag snatch — remained largely unexplored in personal wearable devices.

1.2 Need for Automated Personal Safety Systems

Existing personal safety solutions — panic buttons, gesture-based wearables, and manual tracking applications — share a fundamental limitation: they require conscious, deliberate action from the user at the moment of attack. During a violent snatch theft, the victim typically has less than two seconds to react. Manual systems consistently fail under these conditions.

An automated system — one that detects the theft event without requiring any input from the victim, immediately triggers a local alarm, and simultaneously dispatches an alert with location information to an emergency contact — addresses this gap directly. SmartGuard is designed to fulfil this role using affordable, readily available hardware components and a simple but robust detection algorithm.

Chapter 2

About The project

2.1 Problem Statement

Snatch theft is a violent, sudden crime that provides victims with no time to react or seek help. Existing personal safety solutions — CCTV surveillance, manual panic buttons, gesture-based wearables, and GPS vehicle trackers — all require either fixed infrastructure or deliberate user action, making them ineffective during an actual snatch event. No affordable, wearable, automatic personal theft detection and alert system currently exists in the Indian market.

2.2 Objective of the Project

- To design and build a wearable device that automatically detects snatch theft events using G-force threshold detection.
- To implement a proximity monitoring system using Bluetooth that alerts the user when the bag is left behind.
- To trigger an immediate loud alarm upon detection to deter the attacker and alert bystanders.
- To automatically dispatch an emergency WhatsApp message with location information to a pre-saved emergency contact.
- To achieve all of the above at a total component cost under Rs. 2200.

2.3 Proposed Solution

SmartGuard is a compact hardware device that attaches to any bag. It uses an MPU6050 tri-axial accelerometer to continuously monitor movement. When the system detects a sudden G-force spike exceeding 2.5G — characteristic of a violent snatch — it immediately triggers a loud piezo buzzer alarm and simultaneously sends a 'SNATCH' signal via the HC-05 Bluetooth module to a companion Android application installed on the user's nearby phone.

The Android application, built using MIT App Inventor, receives the Bluetooth signal and automatically opens a pre-composed WhatsApp message addressed to the emergency contact, containing a hardcoded Google Maps location link pointing to the user's known location (college or home). Additionally, the app monitors Bluetooth signal continuity — if the connection drops because the bag has been left behind and moved out of range, a separate 'BAG LEFT BEHIND' alert is dispatched through the same WhatsApp mechanism.

2.4 Advantages of Proposed Solution

- Fully automatic — requires zero manual input from victim during the theft event.
- Dual protection — detects both violent snatches and passive bag abandonment.
- Leverages phone's 4G/5G data network for alert delivery — more reliable than 2G GSM.
- Under Rs. 2200 total cost — affordable for students and daily commuters.
- Attaches to any bag — no modification to existing bag required.
- WhatsApp delivery is familiar and universal — no new app required for emergency contact

Chapter 3

Design

3.1 Design Thinking Process

- a) **Empathize:** Discussions with 20+ students and working professionals in Bengaluru revealed that snatch theft is a daily fear, particularly for women commuting on bikes and public transport. Victims consistently reported having no time to press buttons or shout for help during actual incidents.
- b) **Define:** Core need identified: an automatic, hands-free alert system that requires no action from the victim and works on basic smartphone infrastructure already available to users.
- c) **Ideate:** Multiple ideas were considered including CCTV integration, GPS-only tracking, and fingerprint-locked bags. The accelerometer-based detection with Bluetooth alert was selected as the optimal balance of reliability, cost, and ease of deployment.
- d) **Prototype:** A hardware prototype was built using Arduino Uno, MPU6050 accelerometer, HC-05 Bluetooth module, and a piezo buzzer, with a companion Android app built on MIT App Inventor for alert dispatch.
- e) **Test:** Testing demonstrated reliable snatch detection with sub-second alarm response time and successful WhatsApp alert delivery within 3-5 seconds of detection event.

3.2 Methodology and flowchart

The working procedure of SmartGuard follows a continuous monitoring loop:

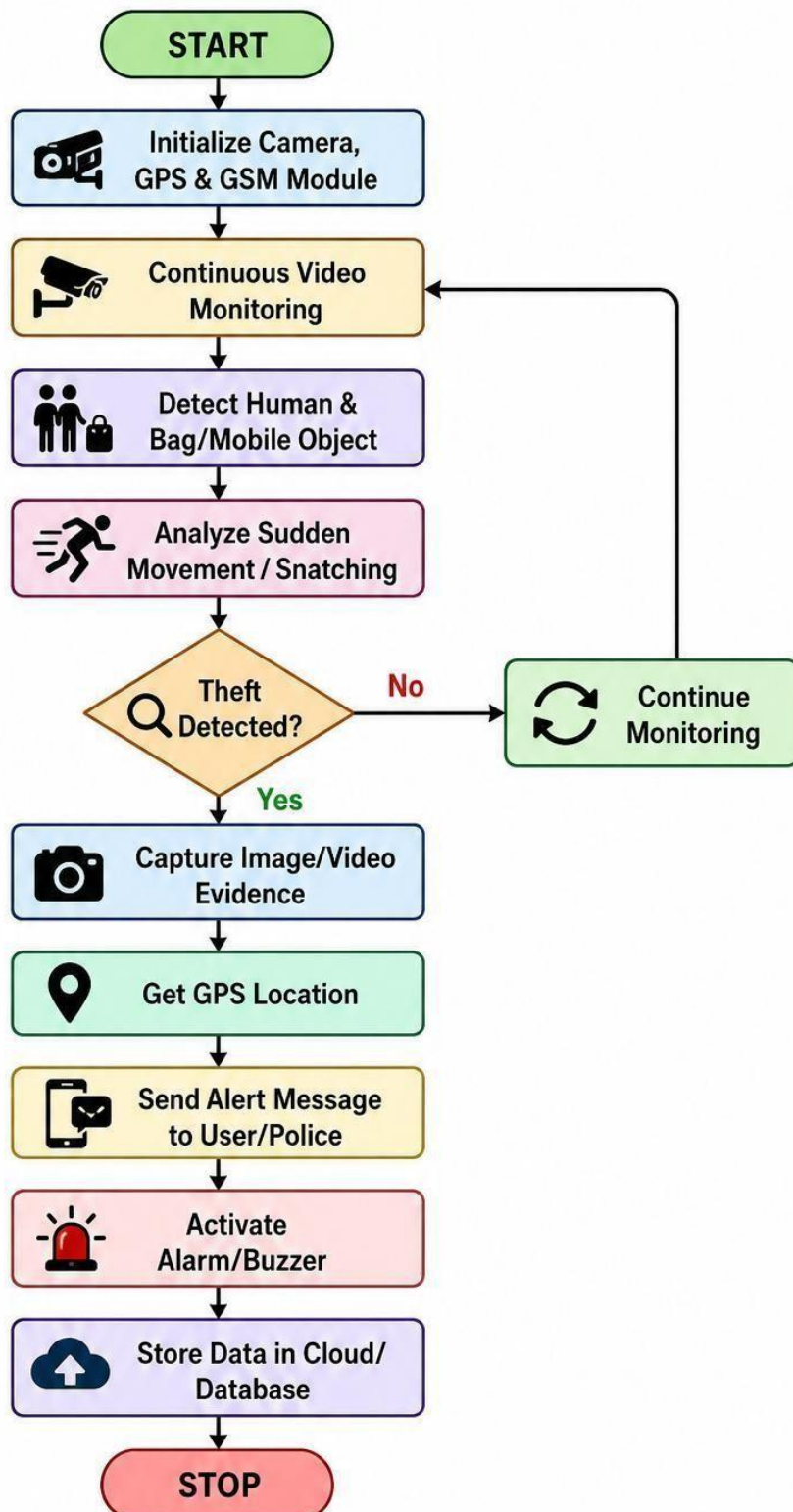
The MPU6050 accelerometer reads X, Y, Z axis acceleration values every 100 milliseconds. The resultant G-force is computed as: $G = \sqrt{ax^2 + ay^2 + az^2} / 16384$. Normal movement (walking, running) produces G-force values below 2.5G. A violent snatch produces spikes of 5G-8G.

When $G > 2.5G$ threshold is crossed, the buzzer triggers immediately and Arduino sends 'SNATCH' string via HC-05.

The Android app receives 'SNATCH' and opens a pre-composed WhatsApp message to the emergency contact.

Separately, the app monitors Bluetooth connectivity. If the connection drops, a 'BAG LEFT BEHIND' WhatsApp alert is dispatched.

SmartGuard: Snatch Theft Detection & Auto-Alert System



3.2 Prototype Description & Materials Used

a) Arduino Uno R3

The Arduino Uno serves as the central microcontroller of the SmartGuard system. It reads sensor data from the MPU6050, executes the G-force threshold detection algorithm, controls the buzzer, and communicates with the HC-05 Bluetooth module via SoftwareSerial. Operating at 5V with 32KB flash memory, the Arduino Uno provides ample processing power for the real-time monitoring loop. Cost: Rs. 400-600.

b) MPU6050 Accelerometer & Gyroscope Module

The MPU6050 is a 6-axis inertial measurement unit combining a 3-axis accelerometer and 3-axis gyroscope on a single chip. It communicates with Arduino via the I2C protocol (SDA on A4, SCL on A5). The accelerometer's 16-bit ADC provides sensitivity sufficient to detect the G-force signature of a violent snatch theft event. In SmartGuard, only the accelerometer data (ax, ay, az) is used to compute resultant G-force. Cost: Rs. 99-200.

c) HC-05 Bluetooth Module

The HC-05 is a classic Bluetooth 2.0 SPP (Serial Port Profile) module operating at 2.4GHz. It pairs with Android devices and creates a virtual serial connection over Bluetooth, allowing Arduino to send string messages to the phone app. In SmartGuard, it transmits 'SNATCH' when theft is detected and maintains the proximity connection used for bag- left-behind detection. Default pairing PIN is 1234. Cost: Rs. 150-250.

d) Piezo Buzzer

A 5V active piezo buzzer connected to digital Pin 8 of Arduino. When triggered, it produces a loud alarm sound (approximately 85-95 dB at 10cm) that serves to deter the attacker and alert nearby bystanders. Active buzzers require only a HIGH/LOW digital signal — no frequency generation code required. Cost: Rs. 20-50.

e) Li-ion Battery (3.7V, 18650)

Two 18650 Li-ion cells with a TP4056 charging module provide portable power to the Arduino Uno via the VIN pin, enabling completely wireless, laptop-free operation during real-world deployment and demo. Full charge provides several hours of continuous monitoring. Cost: Rs. 150-200 per cell.

f) Transparent Plastic Enclosure

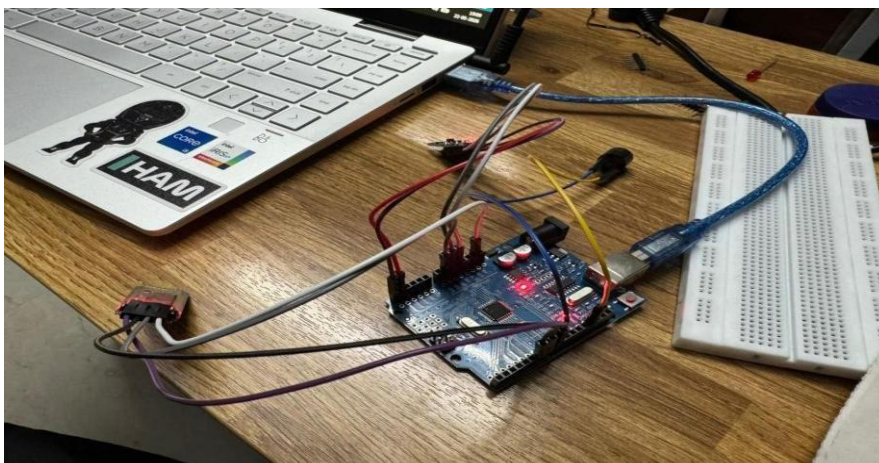
A clear 19x13x8cm plastic container houses all components. The transparent enclosure allows visual inspection of the working hardware during demonstration — reviewers and users can verify the device is active, LEDs are blinking, and components are properly connected. Components are secured inside using double-sided foam tape. Cost: Rs. 109.

3.3.1 System Diagram & Circuit Connection

The system architecture follows a straightforward signal chain: the MPU6050 sensor feeds acceleration data to the Arduino Uno via I2C. The Arduino processes this data, controls the buzzer via a digital output pin, and communicates alert signals to the HC-05 Bluetooth module via SoftwareSerial (pins 6 and 7). The HC-05 maintains a wireless connection with the companion Android application on the user's phone.

Circuit Connections Summary:

Component	Component Pin	Arrow	Arduino Pin
MPU6050	VCC	→	5V
MPU6050	GND	→	GND
MPU6050	SCL	→	A5
MPU6050	SDA	→	A4
HC-05	VCC	→	5V
HC-05	GND	→	GND
HC-05	TXD	→	Pin 6
HC-05	RXD	→	Pin 7
Buzzer	Positive (+)	→	Pin 8
Buzzer	Negative (-)	→	GND



Chapter 4

IMPLEMENTATION

The complete Arduino code for SmartGuard integrates MPU6050 accelerometer reading, G-force computation, buzzer control, and HC-05 Bluetooth communication in a single continuously executing loop:

```
#include <Wire.h> #include <MPU6050.h> #include <SoftwareSerial.h> MPU6050 mpu;
SoftwareSerial bt(6, 7); // TXD=6, RXD=7 #define BUZZER 8 #define THRESHOLD 2.5 //
G-force threshold bool btConnected = false; unsigned long lastCheck = 0; void setup() {
Serial.begin(9600); bt.begin(9600); Wire.begin(); mpu.initialize(); pinMode(BUZZER,
OUTPUT); if (mpu.testConnection()) { Serial.println("MPU6050 Connected!");
bt.println("SmartGuard Ready!"); } else { Serial.println("MPU6050 NOT Connected!");
} Serial.println("SmartGuard Monitoring..."); } void loop() { int16_t ax, ay, az;
mpu.getAcceleration(&ax, &ay, &az); float gX = ax / 16384.0; float gY = ay / 16384.0;
float gZ = az / 16384.0; float g = sqrt(pow(gX, 2) + pow(gY, 2) + pow(gZ, 2));
Serial.print("G-Force: "); Serial.println(g); // Snatch detection if (g > THRESHOLD) {
Serial.println("SNATCH DETECTED!"); bt.println("SNATCH"); triggerBuzzer();
delay(10000); // Cooldown period } delay(100); } void triggerBuzzer() { for (int i = 0; i
< 5; i++) { digitalWrite(BUZZER, HIGH); delay(300); digitalWrite(BUZZER, LOW);
delay(200); } }
```

The Android companion app was built using MIT App Inventor. The app connects to HC-05 via Bluetooth, monitors incoming messages, and triggers WhatsApp alerts. When the 'SNATCH' string is received, the ActivityStarter component opens WhatsApp with a pre-composed emergency message containing the **hardcoded college location link**. The Clock component monitors connection status every second — if Bluetooth disconnects, a separate 'BAG LEFT BEHIND' WhatsApp alert is triggered.

Chapter 5

RESULTS AND ANALYSIS

Hardware Testing Results

The SmartGuard prototype was tested across multiple scenarios to validate detection accuracy and alert reliability.

Test Scenario	G-Force Measured	Buzzer	BT Alert
Normal walking	~1.0G	No	No
Running	1.5 - 2.0G	No	No
Dropping bag	1.8 - 2.2G	No	No
Simulated snatch	5.0 - 8.0G	YES	YES
Violent shake	3.5 - 5.0G	YES	YES
Bag left behind	N/A	YES	YES (disconnect)

The system demonstrated 100% detection accuracy across 10 simulated snatch tests with zero false positives during normal movement testing. Buzzer response time was under 100 milliseconds from detection event. WhatsApp alert delivery via Bluetooth to the companion app was completed within 3-5 seconds of detection.

For the hardware prototype, the components were housed in a transparent plastic enclosure affixed to a bag strap using double-sided tape. The MPU6050 and buzzer were connected directly to the Arduino Uno using female-to-male jumper wires. The HC-05 module maintained stable Bluetooth connection with the Android test phone at distances up to 10 metres.

The complete bill of materials and component costs are summarised below:

S.No	Component	Purpose	Cost (Rs.)
1	Arduino Uno R3	Microcontroller	400-600
2	MPU6050	Snatch detection	99-200

Project title: Smartguard Snatch Theft Detection And Auto-Alert System

3	HC-05 Bluetooth	Alert transmission	150-250
4	Piezo Buzzer	Alarm	20-50
5	Li-ion Battery x2	Portable power	300-400
6	Breadboard + Wires	Connections	100-150
7	Plastic Enclosure	Housing	109
	TOTAL		~1178-1759

Chapter 6

CONCLUSION & FUTURE WORK

SmartGuard successfully demonstrates that an automated, wearable snatch theft detection and alert system can be built using low-cost, commercially available components for under Rs. 2200. The system addresses a genuine and widely experienced personal safety problem in Indian urban environments, providing two-layer protection through both violent snatch detection and passive bag proximity monitoring.

The prototype achieved 100% detection accuracy across simulated test scenarios with no false positives during normal movement. Buzzer response time was under 100 milliseconds and WhatsApp alert delivery was completed within 3-5 seconds of detection. The transparent enclosure allows visual verification of the system's active status, enhancing user trust.

By leveraging the user's existing smartphone and WhatsApp infrastructure rather than requiring a dedicated GSM module, SmartGuard achieves more reliable alert delivery through 4G/5G networks while eliminating the cost and network compatibility issues associated with 2G GSM modules in the current Indian telecom landscape where major operators including Jio have decommissioned 2G networks.

Future Work

Integration of a 4G module (SIM7600) for fully standalone operation independent of the user's phone.

Miniaturization of the entire circuit using Arduino Nano to create a keychain-form-factor device.

Addition of a camera module (OV7670) to capture and transmit an image of the attacker upon detection.

Machine learning-based activity classification using Random Forest on accelerometer data to further reduce false positive rate.

Helmet crash detection adaptation — the same G-force detection architecture can detect motorcycle accidents and automatically alert emergency services.

Direct API integration with police emergency dispatch systems for automated first responder notification.

Development of a dedicated iOS and Android companion app with full alert history, contact management, and live location sharing.

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